

Indian Statistical Institute
B. Math. Hons. III Year
Semestral Examination 2010-2011
Differential Geometry

Duration: 3 Hours

Marks: 100

Instructor: B. Bagchi

1. (a) Which of the following three curves are regular?
(i) $t \mapsto (\cos^2 t, \sin^2 t)$, $-\infty < t < \infty$,
(ii) $t \mapsto (\cos^2 t, \sin^2 t)$, $0 < t < \pi/2$,
(iii) $t \mapsto (t, \cosh t)$, $-\infty < t < \infty$.
(b) Find unit speed reparametrisations of the regular curves among these. [6 + 9 = 15]
2. If γ is a unit speed curve in $\mathbb{R}^3$ with constant curvature (non-zero) and zero torsion then show that γ is an arc of a circle. [15]
3. (a) Deduce the Frenet-Serret equations for a unit speed curve in $\mathbb{R}^3$ with no-where vanishing curvature.
(b) Show that for any two positive smooth functions K and T on $\mathbb{R}$, there is a unit speed curve in $\mathbb{R}^3$ with curvature K and torsion T . [10 + 15 = 25]
4. Let $U_1 = (-1/2, 1/2) \times (0, 2\pi)$ and $U_2 = (-1/2, 1/2) \times (-\pi, \pi)$. Define $\sigma_i : U_i \rightarrow \mathbb{R}^3$ by $\sigma_i(t, \theta) = ((1 - t \sin \frac{\theta}{2}) \cos \theta, (1 - t \sin \frac{\theta}{2}) \sin \theta, t \cos \frac{\theta}{2})$ $i = 1, 2$. Show that $\{\sigma_1, \sigma_2\}$ is an atlas for a smooth surface S . Compute the unit normal to S along the “meridian circle” corresponding to $t = 0$. Hence prove that S is not orientable. [5 + 10 + 5 = 20]
5. Let K_1 and K_2 be the principal curvatures of a co-ordinate chart σ for a surface. Show that:
(a) K_1, K_2 are real numbers.
(b) If $K_1 = K_2$ at some point then the two fundamental forms at that point are scalar multiples of each other,
(c) if $K_1 \neq K_2$ at some point then corresponding principal vectors at that point are orthogonal. [10 + 8 + 7 = 25]